Project Proposal (Proposed Solution) Report

Date	13 November 2026
Team ID	0157CS241131
Project Name	AI Travel Itinerary Planner (Trip Genius AI)
Maximum Marks	5 Marks
Project Title	AI Travel Itinerary Planner

This proposal is about making trip planning easier with an AI model. The app creates a day-wise itinerary from the traveller's choices and lets the user save and export it.

Project Overview

Objective	Reduce the time spent on planning a trip by generating an itinerary from the destination, dates, interests, budget and pace using a Large Language Model.
Scope	Planner form, AI generation, day-wise display, Supabase authentication, save/manage trips, PDF export, cloud deployment. Booking, payments, live prices and GPS navigation are outside scope.

Problem Statement

Description	Travellers spend hours collecting information from many websites and still do not get a proper schedule that matches their needs.
Impact	A solution saves the traveller's time, keeps the plan within budget and gives one place to store and download plans.

Proposed Solution

Approach	The React frontend collects the preferences. FastAPI checks them and builds the prompt for Groq (openai/gpt-oss-120b). The reply is parsed and checked, and Supabase stores the trips with row level security.
Key Features	• AI itinerary with Morning / Afternoon / Evening activities, transport, food and cost hints • Budget tiers and pace control • Secure sign-up/login and private My Trips • Rename, delete and reopen trips • PDF export

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		

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Computing Resources	Development machine	Any modern laptop/PC (no GPU needed; AI runs on Groq cloud)
Memory	RAM	8 GB recommended (4 GB minimum)
Storage	Disk	2 GB free for Node, Python and project files
Software		
Frameworks	Frontend / Backend	React 19 + Vite 6; FastAPI
Libraries	Additional	Pydantic, httpx, groq SDK, supabase-js, jsPDF, pytest
Development Environment	IDE	VS Code / Antigravity; Git and GitHub
Cloud Services	Hosting and data	Vercel, Render, Supabase, Groq accounts
Data		
Data	Source, size, format	No static dataset; runtime user input (JSON) and LLM output (JSON); trips and itineraries stored in PostgreSQL